Mathematical methods

The document formerly known as "'the' calculus"

ME

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To add: multipoles and spherical harmonics, big O and asymptotic notation and thms, see "Representation of a source-free vector field as a curl" Ch V appendix 2 in Courant vol ii, calculus of variations (courant vol 2?), complex stuff (courant?), integrating factors (for exact differentials + more?), sturm-liouville eval problems + (higher dim generalization?), perturbation theory

lame coefficients...

$$\nabla^2 = g^{mn}\left(\partial_m\partial_n - \Gamma^l_{mn}\partial_l\right) \tag{1}$$

$$= \frac{1}{\sqrt{\det g}}\partial_i\left(\sqrt{\det g}g^{ij}\partial_j\right) \tag{2}$$

1 Differential equations

(associated) legendre, bessel, spherical harmonics +addition theorem

1.1 Poisson equation

$$\nabla^2 u(\mathbf{x}) = -\rho(\mathbf{x}) \tag{3}$$

1.2 Screened Poisson equation

$$\left[\nabla^2 - \lambda^2\right] u(\mathbf{x}) = \rho(\mathbf{x}) \tag{4}$$

1.3 Navier-Stokes equations

1.3.1 Stokes equations (Low Reynolds number)

$$-\boldsymbol{\nabla} p(\mathbf{x} + \eta \nabla^2 \mathbf{v}(\mathbf{x}) + \mathbf{f}(\mathbf{x}) = 0 \tag{5}$$

$$\boldsymbol{\nabla} \cdot \mathbf{v}(\mathbf{x}) = 0 \tag{6}$$

1.4 Sturm-Liouville theory

Given functions $p(x), q(x)$ and <u>weight</u> $w(x) > 0$, find eigenvalue λ and eigenfunction $u(x)$ satisfying

$$(p(x)u'(x))' + q(x)u(x) + \lambda w(x)u(x) = 0 \tag{7}$$

subject to some boundary conditions (e.g. $u(a) = 0 = u(b)$). If above things satisfy: $p(x), w(x) > 0$; p, p', q, w are continuous on $[a, b]$; boundary conditions are given by

$$\alpha_1 u(a) + \alpha_2 u'(a) = 0 \qquad \alpha_1^2 + \alpha_2^2 > 0 \tag{8}$$

$$\beta_1 u(b) + \beta_2 u'(b) = 0 \qquad \beta_1^2 + \beta_2^2 > 0 \tag{9}$$

the problem is said to be regular and we can say some nifty stuff.

- eigenvalues λ_n are countable, can be indexed so as to be strictly increasing $\lambda_n < \lambda_{n+1}$, and are unbounded $\lambda_n \to \infty$
- corresponding (normalized)eigenfunctions $u_n(x)$ have exactly $n-1$ zeros in $[a, b]$ and are orthonormal wrt the weighted inner product

$$\int_a^b u_m(x) u_n(x) w(x) \, \mathrm{d}x = \delta_{mn} \tag{10}$$

Define the differential operator $Lu = (pu')' + qu$ so s-l problem is

$$\big(Lu\big)(x) = -\lambda w(x) u(x). \tag{11}$$

and if u, v satisfy the regular bc's above we have

$$\int_a^b vLu \, \mathrm{d}x = \int_a^b (Lv)u \, \mathrm{d}x \, . \tag{12}$$

Equivalent variational problem: define

$$I[u] = \frac{\int_a^b p(x)(u'(x))^2 - q(x)u(x)^2 \, \mathrm{d}x}{\int_a^b u(x)^2 w(x) \, \mathrm{d}x} \tag{13}$$

If u_0 minimizes $I[u]$

$$\delta I[u_0, \delta u] = 0 \qquad \forall \delta u \text{ with } \delta u(a) = \delta u(b) = 0 \tag{14}$$

then u_0 solves the eigenvalue problem with $\lambda = I[u_0]$

$$\big(Lu_0)(x) = -I[u_0]w(x)u(x) \tag{15}$$

1.5 Associated Laguerre equation

For $\lambda, \nu \in \mathbb{R}$

$$xy'' + (\nu + 1 - x)y' + \lambda y = 0 \tag{16}$$

Solutions $U(-\lambda, 1 + \nu, x)$ where U is a confluent hypergeometric function of the first kind and when $\lambda = n = 0, 1, 2, 3, ...$ associated Laguerre polynomials $L_n^\nu(x)$.

- comes up in solutions to schrodinger equation. U solutions for quantum defect or something. something something $1/r$ potentials...

1.5.1 Laplace's Equation, Associated Legendre Polynomials, and Spherical Harmonics

Laplace's equation

$$\nabla^2 \Phi = 0 \tag{17}$$

describes in a region where $\rho = 0$. In spherical coordinates, the general solution (obtained by separation of variables) is given by

$$\Phi(\mathbf{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \left[A_{lm} r^l + B_{lm} r^{-(l+1)} \right] Y_{lm}(\theta, \phi) \tag{18}$$

where A_{lm} and B_{lm} are chose to satisfy boundary conditions and Y_{lm} are spherical harmonics.

1.6 Symmetries of differential equations and Lie groups

2 vector calc

every (real) linear transformation is a combo of rotation, skew, reflection, and scaling, and every function is locally linear, SOOOOO....helmholtz theorem? curl is a rotaion, divergence is scaling...

how does curl depend on orientation?

f(x) as a transformation: maps number line to number line. the derivative is a measure of the degree of squishing/stretching! negative derivative means stuff is flipped/reflected

A one-dimensional Riemannian manifold does not have any intrinsic curvature at all. It is always locally isometric to a straight, "flat" line.

Formally, the Riemann curvature tensor has but a single component R1111 , but this element is required to be 0 due to (for example) the skew-symmetry of Rijkl.

for 1d embedded submanifold (space curve) the extrinsic curvature (curvature and torsion of the curve) and the metric ($\mathbf{r}' \cdot \mathbf{r}' = s'^2$) determine the embedding. is this true for larger dimensional submanifolds? e.g. does Gaussian curvature tensor + metric give embedding?

The right definition is that det(A) is the volume of the image of the unit cube after applying the transformation determined by A. its like the amount by which linear transformation scales volumes, the sign tells you if it switches orientation of the space

can you think of volume integral as a thing 'doted' with a differential vectorish thing, but the 'differential volume vector' lives in a one dimensional space since its a 3-form.

orientation choice of V is equivalent to choosing a basis for the (one-dimensional) space of top forms (dim(V)-forms) on V

volume form is just a smooth choice of non-zero top forms at tangent spaces! (i.e. a smooth basis for topforms) so this gives you orientation + (extra bits)

Any mathematization of space-space-space must make sense of all the things that stuff in space can do. Things can occupy locations (you need linear positions, you can point to where the thing is), things can face directions (you need angular positions), things can change locations by some amount in some direction (you need lengths and directions), and things can change how they face by some amount in some sense (you need angles='angular lengths' and orientation='angular directions').

One approach is to use a Riemannian-ish manifold (M, g, ε) with a volume form (orientation for tangent spaces): treat locations in space as points in the manifold, identify 'angular positions' with (unit) tangent vectors[1], treat directions you can move in as tangent vectors, use the metric to measure lengths and 'angular lengths', and use the volume form to make sense of angular orientations.

Can we also use 'linear' position manifold with 'linear' metric along with 'angular' position manifold with 'angular metric' $(M, g) \times (N, h)$. For Newtonian space, use 3d Euclidean space $M = \mathbb{E}^3$ and 3d rotation group $N = SO(3)$. Then, an identification of the tangent spaces gives you a volume form!!!!!! This identification is like defining the hat map to associate each $T_x M$ with $\mathfrak{so}(3) = T_I N$. Can we think of this as coming from a choice of (smoothly varying) representation of $SO(3)$ on each $T_p M$

The dot product of $\mathbf{u}, \mathbf{v}$ is the scalar defined by:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos\theta \tag{19}$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

[1] does the arbitrary right/left-handed $\pm$ identification come from the 2-1 mapping of unit vectors to axes of rotation?

The cross product of $\mathbf{u}, \mathbf{v}$ is the unique vector with the following properties:

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta \tag{20}$$

$$\mathbf{u} \times \mathbf{v} \perp \mathbf{uv} \tag{21}$$

$$sgn(\mathbf{u}, \mathbf{v}, \mathbf{u} \times \mathbf{v}) > 0 \tag{22}$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

The special orthogonal group $SO(3)$ is the group of rotations about a fixed point (origin) in three dimensional space.

a rotation can be specified by the angle of rotation ϕ and the axis about which things rotate $\mathbf{n}$.

3 crazy shit from buckling notes too dumb for git

enforcing constraints and gauge field

Let $\mathscr{L}_0$ be the some Lagrangian, suppose you want to enforce some constraint or just rewrite the lagrangian in a more convenient form (e.g. get rid of pesky square roots). Pull a new dynamical degree of freedom e out of nowhere and come up with a new e-dependent action in such a way that the Euler-Lagrange equation obtained from varying e is an algebraic relationshiup between

The differential arclength $\mathrm{d}s = s' \,\mathrm{d}\xi$ is the invariant 1-dimensional volume element since $(s')^2 = \mathbf{r}' \cdot \mathbf{r}'$ is the metric induced by the curve's embedding in $\mathbb{E}^3$. In order to write the energy density $\mathscr{E}$ in a...

- $$\mathscr{E}\,\mathrm{d}s = \mathscr{E}\sqrt{\mathbf{r}' \cdot \mathbf{r}'}\,\mathrm{d}\xi \mapsto \left(\frac{1}{2e}\mathbf{r}' \cdot \mathbf{r}' + \frac{e}{2}\right)\mathscr{E}\,\mathrm{d}\xi \tag{23}$$

  Then the Euler-Lagrange equation obtained from varying e is

  $$(-\frac{1}{2e^2}\mathbf{r}' \cdot \mathbf{r}' + \frac{1}{2})\mathscr{E} = 0 \Rightarrow e^2 = \mathbf{r}' \cdot \mathbf{r}' \tag{24}$$

  Include the constraint $\mathbf{r}' = s'\mathbf{d}_1$ by adding $\int \mathbf{f} \cdot (\mathbf{r}' - e\mathbf{d}_{(1)})\,\mathrm{d}\xi$ to the action. The Euler-Lagrange equation obtained from varying e then becomes

  $$(-\frac{1}{2e^2}\mathbf{r}' \cdot \mathbf{r}' + \frac{1}{2})\mathscr{E} - \mathbf{d}_{(1)} \cdot \mathbf{f} = 0 \tag{25}$$

  which, after applying the constraint $\mathbf{r}' = s'\mathbf{d}_1$ gives

  $$\mathbf{d}_{(1)} \cdot \mathbf{f} = 0 \tag{26}$$

- You can also use

  $$\mathscr{E}\,\mathrm{d}s = \mathscr{E}\sqrt{\mathbf{r}' \cdot \mathbf{r}'}\,\mathrm{d}\xi \mapsto \left(\frac{1}{2e}\mathbf{r}' \cdot \mathbf{r}' + \frac{e}{2}\mathscr{E}^2\right)\mathrm{d}\xi \tag{27}$$

  Then the Euler-Lagrange equation obtained from varying e is

  $$-\frac{1}{2e^2}\mathbf{r}' \cdot \mathbf{r}' + \frac{1}{2}\mathscr{E}^2 = 0 \Rightarrow e^2 = \frac{\mathbf{r}' \cdot \mathbf{r}'}{\mathscr{E}^2} \tag{28}$$

4 Approximation methods

4.1 Shooting method

Solves boundary value problem by guessing initial values and updating until boundary conditions are satisfied.

5 Continuity and conservation laws

Sometimes the amount of stuff in a region changes. Sometimes it changes because the stuff went somewhere else and sometimes it changes because the stuff was created or destroyed.

5.1 Continuity equation

Let q be the amount of 'stuff' in region V. Suppose with density ρ_q, flux density (current) j_q, and creation rate per unit volume r_q. Then, for any fixed volume Ω the amount of q in Ω is changing at a rate given by the sum of the rate at which q is entering from the outside (i.e. crossing $\partial\Omega$) and the rate at which q is being created inside Ω.

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho_q \,\mathrm{d}^3x = -\oint_{\partial\Omega} j_q \cdot \mathrm{d}\mathbf{S} + \int_\Omega r_q \,\mathrm{d}^3x \tag{29}$$

$$= -\int_{\partial\Omega} \boldsymbol{\nabla}\cdot j_q \,\mathrm{d}^3x + \int_\Omega r_q \,\mathrm{d}^3x\,. \tag{30}$$

Since the region was arbitrary, the above integral implies a local continuity equation.

$$\frac{\partial \rho_q}{\partial t} + \boldsymbol{\nabla}\cdot j_q = r_q \tag{31}$$

Note that integrating over material volumes results in the same equation.

Conservation of mass: $\rho_q \mapsto \rho$, $j_q \mapsto \rho\mathbf{v}$, $r_q \mapsto 0$[2]

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\mathbf{v}) = 0 \tag{32}$$

Momentum: $\rho_q \mapsto \rho\mathbf{v}$, $j_q \mapsto \rho\mathbf{v}\mathbf{v}$, $r_q \mapsto \mathbf{f} + \boldsymbol{\nabla}\cdot\sigma$.

$$\mathbf{f} + \boldsymbol{\nabla}\cdot\sigma = \frac{\partial}{\partial t}(\rho\mathbf{v}) + \boldsymbol{\nabla}\cdot(\rho\mathbf{v}\mathbf{v}) \tag{33}$$

$$= \mathbf{v}\Big[\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\mathbf{v})\Big] + \rho\frac{\partial\mathbf{v}}{\partial t} + \rho\mathbf{v}\cdot\boldsymbol{\nabla}\mathbf{v} \tag{34}$$

$$= \rho\frac{\partial\mathbf{v}}{\partial t} + \rho\mathbf{v}\cdot\boldsymbol{\nabla}\mathbf{v} \tag{35}$$

[2] unless there is some source of additional mass or whatevs

6 Curves, surfaces, and tensors

Let's take the position vector $\mathbf{r}$, length element $\mathrm{d}s^2$, and angles to be the fundamental things. From these we can define an inner (dot) product. Once a coordinate system (x^1, x^2, x^3) is chosen we can define co/contravariant bases and the components of the metric tensor.

$$\mathrm{d}s^2 = g_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j \qquad \mathbf{e}_i := \frac{\partial \mathbf{r}}{\partial x^i} = \partial_i \mathbf{r} \qquad g_{ij} := \mathbf{e}_i \cdot \mathbf{e}_j \tag{36}$$

$$\mathbf{e}_i \cdot \mathbf{e}^j = \delta_i^j \qquad g_{ik} g^{kj} = \delta_i^j \qquad x_i = g_{ij} x^j \tag{37}$$

Note that in general $\mathbf{r} \neq x^i \mathbf{e}_i$, but only in cartesian coordinate systems it does.

coordinate changes and transformation laws

New coordinates $x^{i'}$ are nice functions $x^{i'} = x^{i'}(x^i)$ of old coordinates x^i. The associated Jacobian determinant must be nonzero in the region of interest

$$\frac{\partial x'}{\partial x} := \det \begin{pmatrix} \frac{\partial x^{1'}}{\partial x^1} & \cdots & \frac{\partial x^{1'}}{\partial x^n} \\ \cdots & \cdots & \cdots \\ \frac{\partial x^{n'}}{\partial x^1} & \cdots & \frac{\partial x^{n'}}{\partial x^n} \end{pmatrix} \neq 0 \tag{38}$$

so that an inverse transformation $x^i = x^i(x^{i'})$ exists. Contravariant and covariant objects can be defined as (the coeficients of) linear combinations of covariant and contravariant bases respectively, or they can be defined by their transformation properties. Things that transform like basis vectors

$$\mathbf{e}_{i'} = \frac{\partial \mathbf{r}}{\partial x^{i'}} = \frac{\partial \mathbf{r}}{\partial x^i}\frac{\partial x^i}{\partial x^{i'}} = \mathbf{e}_i \frac{\partial x^i}{\partial x^{i'}} \tag{39}$$

are called covariant and get subscripts

$$B_{i'} = \frac{\partial x^i}{\partial x_{i'}} B_i. \tag{40}$$

Contravariant vectors get superscripts and transform in the opposite way

$$A^{i'} = \frac{\partial x^{i'}}{\partial x^i} A^i. \tag{41}$$

Every thing is evaluated at whatever point P (in appropriate coordinates) you're interested in.

differentials and Christoffel symbols

$$\mathbf{A} = A^i \mathbf{e}_i = A_i \mathbf{e}^i \tag{42}$$

$$\mathrm{d}\mathbf{A} = \mathrm{d}A^i\,\mathbf{e}_i + A^j\,\mathrm{d}\mathbf{e}_j \tag{43}$$

$$= \left(\mathrm{d}A^i + \Gamma^i_{jk} A^j\,\mathrm{d}x^k\right)\mathbf{e}_i \tag{44}$$

$$\mathrm{d}\mathbf{A} = \mathrm{d}A_i\,\mathbf{e}^i + A_j\,\mathrm{d}\mathbf{e}^j \tag{45}$$

$$= \left(\mathrm{d}A_i - \Gamma^j_{ik} A_j\,\mathrm{d}x^k\right)\mathbf{e}^i \tag{46}$$

where the Christoffel symbol is defined by $\mathrm{d}\mathbf{e}_j = \Gamma^i_{jk}\mathbf{e}_i\,\mathrm{d}x^k = \frac{\partial \mathbf{e}_j}{\partial x^k}\,\mathrm{d}x^k = \frac{\partial \mathbf{r}}{\partial x^j x^k}\,\mathrm{d}x^k$.
Proof:

$$0 = \mathrm{d}\left(\mathbf{e}^i \cdot \mathbf{e}_j\right) = \mathrm{d}\mathbf{e}^i \cdot \mathbf{e}_j + \mathbf{e}^i \cdot \left(\Gamma^l_{jk}\mathbf{e}_l\,\mathrm{d}x^k\right) = \mathrm{d}\mathbf{e}^i \cdot \mathbf{e}_j + \Gamma^i_{jk}\,\mathrm{d}x^k \tag{47}$$

$$\Rightarrow \mathrm{d}\mathbf{e}^i = -\Gamma^i_{jk}\mathbf{e}^j\,\mathrm{d}x^k \tag{48}$$

□

Note that Γ^i_{jk} by itself is not a tensor since it has a funny transformation rule. Define the absolute/covariant differential to be the components of $\mathrm{d}\mathbf{A}$

$$DA^i := \mathrm{d}A^i + \Gamma^i_{jk}A^j\,\mathrm{d}x^k \qquad\qquad DA_i := \mathrm{d}A_i - \Gamma^j_{ik}A_j\,\mathrm{d}x^k \tag{49}$$

and define the covariant derivative as

$$\boldsymbol{\nabla}_j A^i := \frac{\partial A^i}{\partial x^j} + \Gamma^i_{jk}A^k \qquad\qquad \boldsymbol{\nabla}_j A_i := \frac{\partial A_i}{\partial x^j} - \Gamma^k_{ij}A_k. \tag{50}$$

Covariant derivatives are metrinillic

$$\boldsymbol{\nabla}_k g_{ij} = 0 \tag{51}$$

some other useful(?) things...

$$\Gamma^k_{ij} = \frac{1}{2}g^{km}\left(\frac{\partial g_{mi}}{\partial x^j} + \frac{\partial g_{mj}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^m}\right) \qquad\qquad \Gamma^j_{ij} = \frac{1}{2g}\frac{\partial g}{\partial x^i} = \frac{1}{\sqrt{g}}\frac{\partial \sqrt{g}}{\partial x^i} \tag{52}$$

The Levi-Civita tensor/volume element are given by

$$\mathrm{d}V = \sqrt{\det(g_{ij}(x))}\,\mathrm{d}x^1\,\mathrm{d}x^2\,\mathrm{d}x^3 = \sqrt{g}\,\mathrm{d}^3x \tag{53}$$

$$\underline{\underline{\varepsilon}} = \sqrt{\det(g_{ij}(x))}\,\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3 \tag{54}$$

hopefully context will make clear what I mean...[3]

6.1 Shape optimization!

Consider a surface described parametrically

$$Z^i = Z^i(S) \tag{55}$$

Now consider the position vector $\mathbf{R}$ as a function of the surface coordinates S^α

$$\mathbf{R} = \mathbf{R}(S) \tag{56}$$

Definition 1. *The (surface) covariant basis vector $\mathbf{S}_\alpha$ is defined to be the partial derivative of the position vector $\mathbf{R}$ with respect to the surface coordinate S^α*

$$\mathbf{S}_\alpha := \frac{\partial \mathbf{R}}{\partial S^\alpha} \tag{57}$$

These vectors span the tangent space of the surface and can be related to the ambient basis vectors $\mathbf{Z}_i$ by partial differentiation.

$$\frac{\partial \mathbf{R}}{\partial S^\alpha} = \frac{\partial \mathbf{R}}{\partial Z^i}\frac{\partial Z^i}{\partial S^\alpha}$$

Definition 2. *The partial derivatives $\frac{\partial Z^i}{\partial S^\alpha}$ of the equation of the surface with respect to the surface coordinates are known as the shift tensor Z^i_α.*

$$Z^i_\alpha := \frac{\partial Z^i}{\partial S^\alpha} \tag{58}$$

The shift tensor gives a compact expression for the surface basis $\mathbf{S}_\alpha$ in terms of the ambient basis $\mathbf{Z}_i$.

$$\mathbf{S}_\alpha = Z^i_\alpha \mathbf{Z}_i \tag{59}$$

Dotting both sides of this equation with a contravariant basis vector $\mathbf{Z}^i$ gives an explicit expression for the components of the shift tensor.

$$Z^i_\alpha = \mathbf{S}_\alpha \cdot \mathbf{Z}^i \tag{60}$$

The shift tensor relates surface quantities to their ambient counterparts. Consider a vector $\mathbf{T}$ in the tangent space. It can be represented as a linear combination of either the surface or ambient basis vectors.

[3] volume form in terms of structure constants?

$$\mathbf{T} = T^\alpha \mathbf{S}_\alpha = T^\alpha Z^i_\alpha \mathbf{Z}_i$$

$$\mathbf{T} = T^i \mathbf{Z}_i$$

The components of $\mathbf{T}$ with respect to these two bases are related by the shift tensor.

$$\mathbf{T} = T^\alpha \mathbf{S}_\alpha = T^\alpha Z^i_\alpha \mathbf{Z}_i$$

$$T^i = T^\alpha Z^i_\alpha \tag{61}$$

Reverse relation $T^\alpha = T^i Z^\alpha_i$ holds as well but we need to define a few more things to show why.

Definition 3. *The components of the surface covariant metric tensor $S_{\alpha\beta}$ are the dot products of the surface covariant basis vectors $\mathbf{S}_\alpha$ and the surface contravariant metric tensor $S^{\alpha\beta}$ is the matrix inverse of $S_{\alpha\beta}$*

$$S_{\alpha\beta} := \mathbf{S}_\alpha \cdot \mathbf{S}_\beta \tag{62}$$

$$S^{\alpha\beta} S_{\beta\gamma} := \delta^\alpha_\gamma \tag{63}$$

The surface metric tensor is symmetric, positive definite and can be used to juggle surface indices, like its ambient counterpart. It allows us to define

6.1.1 The Normal Vector

6.2 Calculus of moving surfaces

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{a(t)}^{b(t)} f(x,t)\,\mathrm{d}x = \int_{a(t)}^{b(t)} \frac{\partial f(x,t)}{\partial t}\,\mathrm{d}x + f(b(t),t)\dot{b}(t) - f(a(t),t)\dot{a}(t) \tag{64}$$

$$= \int_{a(t)}^{b(t)} \frac{\partial f(x,t)}{\partial t}\,\mathrm{d}x + f(x,t)\dot{x}\big|_{x=a(t)}^{x=b(t)} \tag{65}$$

7 Vector calc and geometry stuff

div/grad/curl in different coordinates, lagrange multipliers, what the flux is flux, differential forms and pullback... see 5.5 in vol1 of sternberg, taylors thm and hessian

1 and 3 dimensional Taylor series

$$f(x+\varepsilon) = \exp\left(\varepsilon \frac{\mathrm{d}}{\mathrm{d}x}\right) f(x) \tag{66}$$

$$= \sum_{n=0}^{\infty} \frac{\varepsilon^n}{n!} f^{(n)}(x) \tag{67}$$

$$f(\mathbf{x}+\mathbf{a}) = \exp(\mathbf{a}\cdot\boldsymbol{\nabla}) f(\mathbf{x}) \tag{68}$$

$$= \sum_{n=0}^{\infty} \frac{(\mathbf{a}\cdot\boldsymbol{\nabla})^n}{n!} f(\mathbf{x}) \tag{69}$$

sometimes these converge or whatever.

Definition 4 (Line integral)**.**

$$\int_C f \,\mathrm{d}l := \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)| \,\mathrm{d}t$$

$$\int_C \mathbf{F}\cdot \mathrm{d}\mathbf{l} := \int_C \mathbf{F}\cdot\hat{\mathbf{t}} \,\mathrm{d}l = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t) \,\mathrm{d}t$$

Where $\mathbf{r} : [a,b] \to C$ *is a smooth bijective parametrization of* $C \subset \mathbb{R}^n$*,* f *is real valued,* $\mathbf{F}$ *is vector valued, and* $\hat{\mathbf{t}} := \frac{\mathbf{r}'}{|\mathbf{r}'|}$ *is the unit tangent to the curve.*

Definition 5 (Surface integral)**.**

$$\int_S f \,\mathrm{d}S := \int_T f(\mathbf{r}(s,t)) \left| \frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t} \right| \mathrm{d}s \,\mathrm{d}t$$

$$\int_S \mathbf{F}\cdot \mathrm{d}S := \int_S \mathbf{F}\cdot\hat{\mathbf{n}} \,\mathrm{d}S = \int_T \mathbf{F}(\mathbf{r}(s,t))\cdot \frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t} \,\mathrm{d}s \,\mathrm{d}t$$

Where $\mathbf{r} : T \subset \mathbb{R}^2 \to S$ *is a smooth bijective parametrization of* $S \subset \mathbb{R}^3$*,* f *is real valued,* $\mathbf{F}$ *is vector valued, and* $\hat{\mathbf{n}} := \frac{\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t}}{\left|\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t}\right|}$ *is the unit normal to the surface.*

7.1 Minimization and variational methods

A function f has a <u>local minimum</u> at the point $\mathbf{x}$ if there is some neighborhood of $\mathbf{x}$ in which the function is bounded below by $f(\mathbf{x})$. A necessary condition for $\mathbf{x}$ to be a local minimum is $\mathrm{d}f_{\mathbf{x}} \equiv 0$. A point at which the derivative of f is zero is called a <u>critical point</u> or <u>stationary point</u>. A sufficient condition for a critical point to be a minimum is that the Hessian of f at the point is positive definite. For a local maximum swap positive definite for negative definite.

- positive/negative definite is equivalent to all eigenvalues being positive/negative.
- If Hessian has both positive and negative eigenvalues $\mathbf{x}$ is a saddle point.
- Each $\frac{\partial^2 f}{\partial x_i \partial x_j} > 0$ is necessary, but is only sufficient for min in 1-dimension.
- The definition of critical point is equivalent to the directional derivative being zero in every direction.

Suppose you want to minimize the real valued function f subject to the constraint(s) $\mathbf{g} \equiv 0$. Define the new function

$$L(\mathbf{x}, \lambda) := f(\mathbf{x}) - \lambda \cdot \mathbf{g}(\mathbf{x}) \tag{70}$$
$$= f(\mathbf{x}) - \lambda^k g_k(\mathbf{x}) \tag{71}$$
$$\tag{72}$$

which depends on variables $\mathbf{x}, \lambda$ and find critical points.

$$\frac{\partial L}{\partial x^k} = 0 \qquad \Rightarrow \qquad \boldsymbol{\nabla} f(\mathbf{x}) - \lambda^k \boldsymbol{\nabla} g_k(\mathbf{x}) = 0 \tag{73}$$
$$\frac{\partial L}{\partial \lambda^k} = 0 \qquad \Rightarrow \qquad g_k(\mathbf{x}) = 0 \tag{74}$$

7.2 Integral theorems

fundamental thm of calc

Theorem 1 (Change of variables)**.** *Let $T : V^* \to V$ by $T(\mathbf{u}) = \mathbf{x}(\mathbf{u})$ be a bijective C^1 map between two "nice" subsets of $\mathbb{R}^n$.*

$$\int_V f(\mathbf{x}) \mathrm{d}^n x = \int_{V^*} f(\mathbf{x}(\mathbf{u})) \left| \frac{\partial(x_i)}{\partial(u_j)} \right| \mathrm{d}^n u$$

- *V is region described in x coordinates*
- *V^* is region described in u coordinates*

Theorem 2 (Stokes)**.**

$$\int_S \mathrm{d}\mathbf{S} \cdot \boldsymbol{\nabla} \times \mathbf{F} = \oint_{\partial S} \mathrm{d}\mathbf{l} \cdot \mathbf{F}$$
$$\int_S \mathrm{d}\mathbf{S} \times \boldsymbol{\nabla} f = \oint_{\partial S} \mathrm{d}\mathbf{l}\, f$$

[4]

Theorem 3 (Divergence)**.**

$$\int_V \mathrm{d}V\, \boldsymbol{\nabla} \cdot \mathbf{F} = \oint_{\partial V} \mathrm{d}\mathbf{S} \cdot \mathbf{F}$$
$$\int_V \mathrm{d}V\, \boldsymbol{\nabla} f = \oint_{\partial V} \mathrm{d}\mathbf{S}\, f$$
$$\int_V \mathrm{d}V\, \boldsymbol{\nabla} \times \mathbf{F} = \oint_{\partial V} \mathrm{d}\mathbf{S} \times \mathbf{F}$$

[4] What are assumptions for div,stokes (e.g. simply connected region?)

Theorem 4 (Green's)**.**

$$\int_V \mathrm{d}V \left[f\nabla^2 g + \boldsymbol{\nabla} f \cdot \boldsymbol{\nabla} g\right] = \int_{\partial V} \mathrm{d}\mathbf{S} \cdot f\boldsymbol{\nabla} g$$

$$\int_V \mathrm{d}V \left[f\nabla^2 g - g\nabla^2 f\right] = \int_{\partial V} \mathrm{d}\mathbf{S} \cdot \left[f\boldsymbol{\nabla} g - g\boldsymbol{\nabla} f\right]$$

7.3 Other useful things

Theorem 5 (Path Independence)**.** *A continuous vector field (1-form)* $\mathbf{F}$ *defined on some open (open necessary?) path connected subset of* $\mathbb{R}^n$ *(or manifold M) is path independent iff* $\mathbf{F} = \boldsymbol{\nabla} F$ *for some function* f *(form is exact).*

Theorem 6. *A vector field* $\mathbf{F}$ *defined on (an open?) simply connected region* $V \subset \mathbb{R}^3$ *is path independent iff* $\boldsymbol{\nabla} \times \mathbf{F} = 0$

Proof: In a simply connected region of $\mathbb{R}^3$ I think that every closed path is the boundary of some surface $C = \partial S$.

$$0 = \int_S \mathrm{d}\mathbf{S} \cdot \boldsymbol{\nabla} \times \mathbf{F} = \oint_C \mathrm{d}\mathbf{l} \cdot \mathbf{F}$$

□

Theorem 7 (Helmholtz)**.** *Let* $\mathbf{F}$ *be a vector field defined in some region* $V \subset \mathbb{R}^3$.

$$\mathbf{F} = -\boldsymbol{\nabla}\Phi + \boldsymbol{\nabla} \times \mathbf{A}$$

$$\Phi(\mathbf{r}) = \frac{1}{4\pi}\int_V \mathrm{d}V' \, \frac{\boldsymbol{\nabla}' \cdot \mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} - \frac{1}{4\pi}\oint_{\partial V} \mathrm{d}\mathbf{S}' \cdot \frac{\mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

$$\mathbf{A}(\mathbf{r}) = \frac{1}{4\pi}\int_V \mathrm{d}V' \, \frac{\boldsymbol{\nabla}' \times \mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} - \frac{1}{4\pi}\oint_{\partial V} \mathrm{d}\mathbf{S}' \times \frac{\mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

If V *is either bounded and* $\mathbf{F}|_{\partial V} = 0$ [5]*, or* $\mathbf{F} \to 0$ *faster than* $1/r$ *as* $r \to 0$ *then the expressions for* Φ *and* $\mathbf{A}$ *can be simplified.*

$$\Phi(\mathbf{r}) = \frac{1}{4\pi}\int_V \mathrm{d}V' \, \frac{\boldsymbol{\nabla}' \cdot \mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

$$\mathbf{A}(\mathbf{r}) = \frac{1}{4\pi}\int_V \mathrm{d}V' \, \frac{\boldsymbol{\nabla}' \times \mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

Proof: It's on Wikipedia yo. □

7.4 differential forms

7.5 moving frames

[5] this works, but maybe there are weaker conditions?

8 Representation theory

If G is a compact connected Lie group, then the exponential map is surjective.

8.1 $SU(2)$ and $\mathfrak{su}(2)$

The 2-dimensional special unitary group

$$SU(2) = \left\{ \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} : |a|^2 + |b|^2 = 1 \right\} \tag{75}$$

is the (Lie) group of 2-by-2 complex unitary matrices with determinant +1. Geometrically, it's the same as 3 sphere $S^3 \subset \mathbb{R}^4$, and algebraically it's the same as unit quaternions. Let's define the following thigs because just trust me ok?

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{76}$$

$$\mathcal{I} = -i\sigma_x \qquad \mathcal{J} = -i\sigma_y \qquad \mathcal{K} = -i\sigma_z \tag{77}$$

$$s_x = \tfrac{1}{2}\sigma_x \qquad s_y = \tfrac{1}{2}\sigma_y \qquad s_z = \tfrac{1}{2}\sigma_z \tag{78}$$

$$\tilde{\mathbf{x}} := \mathbf{x} \cdot \sigma \qquad (\text{for } \mathbf{x} \in \mathbb{R}^3) \tag{79}$$

Some useful identities.

$$\sigma_a \sigma_b = \delta_{ab} + i\epsilon_{abc}\sigma_c \qquad \sigma_a\sigma_b\sigma_c = i\epsilon_{abc} \tag{80}$$

$$[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c \qquad \{\sigma_a, \sigma_b\} = 2\delta_{ab} \tag{81}$$

$$\sigma_a^2 = 1 \qquad \tilde{\mathbf{x}}\tilde{\mathbf{y}} = \mathbf{x} \cdot \mathbf{y} + i\widetilde{\mathbf{x} \times \mathbf{y}} \tag{82}$$

that last thing is a lie algebra isomosphism between su2 with commutator and R3 with cross prod Elements in $SU(2)$ are specified[6] by a unit vector $\hat{\mathbf{n}} = (\cos\alpha, \cos\beta, \cos\gamma) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ (given by direction cosines or canonical coordinates on unit sphere)

$$U_{\hat{\mathbf{n}}}(\psi) = \exp\left(i\tfrac{\psi}{2}\tilde{\mathbf{n}}\right) \tag{83}$$

$$= \cos\tfrac{\psi}{2} + i\tilde{\mathbf{n}}\sin\tfrac{\psi}{2} \tag{84}$$

$$= \begin{pmatrix} \cos\frac{\psi}{2} + in_z\sin\frac{\psi}{2} & (in_x + n_y)\sin\frac{\psi}{2} \\ (in_x - n_y)\sin\frac{\psi}{2} & \cos\frac{\psi}{2} - in_z\sin\frac{\psi}{2} \end{pmatrix} \tag{85}$$

The Lie algebra $\mathfrak{su}(2)$ is the real vector space spanned by is_x, is_y, is_z (or any of the equivalent bases described above). Its complexification $\mathfrak{sl}(2) = \mathfrak{su}(2) + i\mathfrak{su}(2)$...

Let $g_a = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\sigma_a$.

$$g_c^\dagger \sigma_a g_c = (\tfrac{1}{\sqrt{2}} - \tfrac{i}{\sqrt{2}}\sigma_c)\sigma_a(\tfrac{1}{\sqrt{2}} + \tfrac{i}{\sqrt{2}}\sigma_c) \tag{86}$$

$$= \epsilon_{abc}\sigma_b \tag{87}$$

[6] Not uniquely since $U_{-\hat{\mathbf{n}}}(-\psi) = U_{\hat{\mathbf{n}}}(\psi)$ unless you restrict $\psi \in [0, 2\pi)$

Every element is conjugate to an exponential of σ_z

$$\exp\left(i\tfrac{\psi}{2}\tilde{\mathbf{n}}\right) = \exp\left(i\tfrac{\psi}{2}g^\dagger\sigma_z g\right) \tag{88}$$

$$= g^\dagger \exp\left(i\tfrac{\psi}{2}\sigma_z\right) g \tag{89}$$

so the character of the fundamental (i.e. spin $1/2$) representation can be computed using only elements of the form $\exp\left(i\frac{\psi}{2}\sigma_z\right)$.

$$\chi^{(1/2)}(\exp\left(i\tfrac{\psi}{2}\sigma_z\right)) = 2\cos\tfrac{\psi}{2} \tag{90}$$

Fundamental rep

8.2 The group of rotations $SO(3)$

euler angles and cayley klein parameters

active rotations in $\mathbb{R}^3$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \quad R_y(\phi) = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix} \quad R_z(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \qquad L_y = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \qquad L_z = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

9 Special Functions

9.1 Associated Laguerre Polynomials

Solutions to associated Laguerre differential equation with $\nu = k \in \mathbb{Z}, \lambda = n = 0, 1, 2, ...$

Rodrigues formula

$$L_n^k(x) = \frac{e^x x^{-k}}{n!} \frac{\mathrm{d}^n}{\mathrm{d}x^n} \left(e^{-x} x^{n+k} \right) \tag{91}$$

$$= (-1)^k \frac{\mathrm{d}^k}{\mathrm{d}x^k} \left[L_{n+k}(x) \right] \tag{92}$$

generating function

$$g(x, z) = \frac{\exp\left(-\frac{xz}{1-z}\right)}{(1-z)^{k+1}} \tag{93}$$

series

when $k = 0$ we call them Laguerre polynomials.

$$L_n(x) = L_n^0(x) \tag{94}$$

Solutions to Laguerre differential equation with $n = 0, 1, 2, ...$

$$xy'' + (1-x)y' + ny = 0 \tag{95}$$

Several equivalent definitions

Contour integral: contour contains origin, but not $z = 1$

$$L_n(z) = \frac{1}{2\pi i} \oint \frac{e^{-zt(1-t)}}{(1-t)t^{n+1}} \, \mathrm{d}t \tag{96}$$

sum

$$L_n(x) = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \binom{n}{k} x^k \tag{97}$$

9.2 Legendre polynomials and associated legendre polynomials

associated

$$(1-x^2)P_l'^m(x) - 2xP_l'^m(x) + (l(l+1) - \tfrac{m^2}{1-x^2})P_l^m(x) = 0 \tag{98}$$

$$P_l^m(x) = (-1)^m (1-x^2)^{m/2} D^m P_l(x) \tag{99}$$

$$= \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} D^{l+m} (x^2-1)^l \tag{100}$$

- alternate convention $P_{lm} = (-1)^m P_l^m$

For each fixed m, the P_l^m's are <u>orthogonal</u> on $[-1, 1]$.

$$\int_{-1}^{1} P_{l'}^m(x) P_l^m(x) \, \mathrm{d}x = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \delta_{l'l} \tag{101}$$

$$P_l^{-m} = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m \tag{102}$$

10 Cool group theory stuff

10.1 flows

Let M be a G-space. A one parameter subgroup[7] $g(t)$ of G induces a flow given by

$$(x,t) \in M \times \mathbb{R} \mapsto y(t) = g(t)x. \tag{103}$$

For each $x \in M$, $y(t) = g(t)x$ is a curve through passing through x at 'time' $t = 0$ with velocity $\mathbf{v}_x = \frac{\mathrm{d}y}{\mathrm{d}t}|_{t=0}$. This gives us a vector field $\mathbf{v} : M \to TM$ defined by $x \in M \mapsto \mathbf{v}_x \in T_xM$. Now, for each $t \in \mathbb{R}$, let's define the linear operator

$$U(t) : L_2(M) \to L_2(M) \tag{104}$$

$$\Big(U(t)f\Big)(x) = f(g(t)x). \tag{105}$$

Depending on how nice the action is [8], each $U(t)$ is a bounded/unitary and the map $t \mapsto U(t)$ is a strongly/uniformly continuous semigroup!

$$||U(t)f||^2 = \int_M |f(g(t)x)|^2 \,\mathrm{d}\mu\,(x) \tag{106}$$

$$= \int_M |f(y)|^2 \,\mathrm{d}\mu\,(g(-t)y) \tag{107}$$

$$= ||U(t)f||^2 \tag{108}$$

Lets say that $\{\psi_k\}$ is a basis for $L_2(M)$

Then $y(t)$

10.2 actions n stuff

A one parameter subgroup of a Lie group G is a smooth map $g : \mathbb{R} \to G$ which satisfies

$$g(t+s) = g(t)g(s) \qquad \forall s,t \in \mathbb{R} \tag{109}$$

$$g(0) = 1_V \tag{110}$$

Theorem 8. *Every one parameter subgroup can be written as an exponential*

$$g(t) = e^{t\Lambda} \tag{111}$$

for $\Lambda = \dot{g}(0)$

Proof:

$$\frac{\mathrm{d}}{\mathrm{d}s}\Big[g(t+s)\Big]\Big|_{s=0} = \frac{\mathrm{d}}{\mathrm{d}s}\Big[g(t)g(s)\Big]\Big|_{s=0}$$

$$\Rightarrow \dot{g}(t) = g(t)\dot{g}(0)$$

$$\Rightarrow g(t) = Ae^{t\dot{g}(0)}$$

and $g(0) = 1_V \Rightarrow A = 1_V$ □

[7] assume the action is has a power series $g(t)x = x + a_1(x)t + a_2(x)t^2 + ...$ and both $x \mapsto g(t)x$ and $t \mapsto g(t)x$ are smooth/continuous/strongly continuous?

[8] action must preserve the volume form/measure or whatever, be all continuous, etc...

Theorem 9. [9][10] *Let G be a Lie group acting on a space V, let $g(t) = e^{t\Lambda}$ be a one parameter subgroup of G, and let f be a real/complex valued function on V.*

$$f(g(t)\mathbf{x}) = e^{t\mathbf{v}(\mathbf{x})\cdot\boldsymbol{\nabla}} f(\mathbf{x}) \tag{112}$$

Where $\mathbf{v}(\mathbf{x})$ is the directional derivative wrt the vector field $\mathbf{v}(\mathbf{x}) = \frac{\mathrm{d}}{\mathrm{d}t}\big[g(t)\mathbf{x}\big]\big|_{t=0}$. If the action is linear then $\mathbf{v}(\mathbf{x}) = \Lambda\mathbf{x}$. The Where $\mathbf{v}(\mathbf{x}) = \frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t}|_{t=0}$ is the vector field tangent to $\mathbf{y}(t) = g(t)\mathbf{x}$ at $t = 0$ for each $\mathbf{x}$. If for each $\mathbf{x}$ we define $\phi^{(\mathbf{x})} : G \to V$ by $\phi^{(\mathbf{x})}(g) = g\mathbf{x}$ then $\mathbf{v}(\mathbf{x}) = \mathrm{d}\phi_e^{(\mathbf{x})}(\dot{g}(0))$ is the pushforward of $\dot{g}(0)$ at the identity.

strongly continuous one-parameter semigroup....

Proof: Let $\mathbf{y}(t) := g(t)\mathbf{x}$. Expanding $f(\mathbf{y}(t))$ in powers of t about $t = 0$ gives

$$f(\mathbf{y}(t)) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \frac{\mathrm{d}^n}{\mathrm{d}t^n}\Big[f(\mathbf{y}(t))\Big]\Big|_{t=0}$$

Let $f^{(n)}(\mathbf{y}(t)) = \frac{\mathrm{d}^n}{\mathrm{d}t^n}\Big[f(\mathbf{y}(t))\Big]$ so (summing repeated indices and letting subscript i denote differentiation wrt ith variable)

$$\begin{aligned} f^{(n)}(\mathbf{y}(t)) &= \frac{\mathrm{d}}{\mathrm{d}t} f^{(n-1)}(\mathbf{y}(t)) \\ &= \dot{y}^i(t) f_i^{(n-1)}(\mathbf{y}(t)) \\ &= \Lambda^i_k g^k_j(t) x^j f_i^{(n-1)}(\mathbf{y}(t)) \\ &= \Lambda^i_k y^k(t) f_i^{(n-1)}(\mathbf{y}(t)) \end{aligned}$$

Since $\mathbf{y}(0) = \mathbf{x}$, evaluating at $t = 0$ results in

$$\begin{aligned} f^{(n)}(\mathbf{x}) &= \Lambda^i_k x^k f_i^{(n-1)}(\mathbf{x}) \\ &= \Big(\Lambda\mathbf{x}\cdot\boldsymbol{\nabla}\Big) f^{(n-1)}(\mathbf{x}) \end{aligned}$$

So, by the magic of induction we have

$$\frac{\mathrm{d}^n}{\mathrm{d}t^n}\Big[f(\mathbf{y}(t))\Big]\Big|_{t=0} = \Big(\Lambda\mathbf{x}\cdot\boldsymbol{\nabla}\Big)^n f(\mathbf{x})$$

and hence

$$\begin{aligned} f(\mathbf{y}(t)) &= \sum_{n=0}^{\infty} \frac{t^n}{n!}\Big(\Lambda\mathbf{x}\cdot\boldsymbol{\nabla}\Big)^n f(\mathbf{x}) \\ &= \sum_{n=0}^{\infty} \frac{\big(t\Lambda\mathbf{x}\cdot\boldsymbol{\nabla}\big)^n}{n!} f(\mathbf{x}) \\ &= e^{t\Lambda\mathbf{x}\cdot\boldsymbol{\nabla}} f(\mathbf{x}) \end{aligned}$$

[9] I think this is showing the relationship between the infinitesimal generator of the acton on V and the infinitesimal generator of the associated regular representation on space of functions on V.

[10] I think I can extend this to include $e^{\int_0^t \Lambda(s)\mathrm{d}s}$ for $[\Lambda(t_1), \Lambda(t_2)] = 0$ and $\Pi_0^t \exp(\Lambda(s)\,\mathrm{d}s)$

$\square$

If the group action isn't linear, then there are a few options. First, you might be able to embed V in some higher dimensional space where the action is linear (e.g translation $\mathbf{x} \Rightarrow \mathbf{x} + t\mathbf{a}$). Second, even if the action isn't linear we still can write it as an exponential of something. Let $\mathbf{y}(t) = g(t)\mathbf{x}$ and $\mathbf{v} := \frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t}|_{t=0}$ then

$$f(\mathbf{y}(t)) = e^{(t\mathbf{v}\cdot\boldsymbol{\nabla})} f(\mathbf{x}). \tag{113}$$

Note that this can be viewed as a sort of generlization of the Taylor expansion.

We can also just say, given the action of G on V $g \cdot \mathbf{x}$, if we have a one parameter subgroup $g(t)$ then we want to find an operator $U(t)$ which acts on functions on V by

$$U(t)f(\mathbf{x}) = f(g(t)\mathbf{x}) \tag{114}$$

$$\Rightarrow (\frac{\partial U}{\partial t} f)(\mathbf{x}) = \boldsymbol{\nabla}_{g(t)\mathbf{x}} f[\frac{\partial}{\partial t}(g(t)\mathbf{x})] \tag{115}$$

$$\Rightarrow (\frac{\partial U}{\partial t} f)(\mathbf{x}) = (\mathbf{v} \cdot \boldsymbol{\nabla} f)(\mathbf{x}) \tag{116}$$

$$\tag{117}$$

What do we mean... if ψ_k is basis for function space

$$U(t)\psi_k = \sum_l c_{lk}(t)\psi_l \tag{118}$$

so pdv wrt time is just derivatives of all the c_{lk}'s.

$$\frac{\partial U}{\partial t}\psi_k = \sum_l \frac{\partial c_{lk}}{\partial t}\psi_l \tag{119}$$

$$\boldsymbol{\nabla} : \mathcal{F}(M) \to T^*M \tag{120}$$

$$\boldsymbol{\nabla}_{\mathbf{x}} f[\mathbf{v}_{\mathbf{x}}] = \mathbf{v}(\mathbf{x}) \cdot \boldsymbol{\nabla} f(\mathbf{x} \tag{121}$$

11 things to be categorized

$O(n)$... Landau symbol $O(t^2)$ denotes an expression in t whose norm is bounded for all sufficiently small t by a constant multiple of t^2

You can find the eigenvalues of a symmetric matrix A by finding stationary values of $\mathbf{x}^T A\mathbf{x}/\mathbf{x}^T\mathbf{x}$

11.1 laplacian

$$\nabla^2\psi = \tfrac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \tfrac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2} \tag{122}$$

$$=: \tfrac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \tfrac{1}{r^2}\nabla_S^2\psi \tag{123}$$

11.2 Dyadic notation

$$\mathbf{v}\cdot = \mathbf{v}^T \tag{124}$$

The dyadic written as

$$T = \mathbf{uw} \qquad T\cdot\mathbf{v} = \mathbf{uw}\cdot\mathbf{v} = \mathbf{uw}^T\mathbf{v} \qquad \mathbf{v}\cdot T = \mathbf{v}\cdot\mathbf{uw} = \mathbf{v}^T\mathbf{uw} \tag{125}$$

12 Stuff below here still really needs some work. Get your shit together bro.

13 Linear algebra

13.1 Tensor and matrix decompositions

A symmetric 3-by-3 matrix/rank 2 tensor T can be decomposed into isotropic $T_{iso} = \frac{1}{3}\operatorname{tr}(T)I$ (I is id matrix) and anisotropic (traceless) $T - T_{iso}$ parts. Physically, the isotropic part represents a direction independent transformation.

13.2 Complexification

> 'The shortest path between two truths in the real domain passes through the complex domain.'
> -Hadamard

If V is an n-dimensional real vector space then its complexification $H = \mathbb{C} \otimes V = \{\alpha v | \alpha \in \mathbb{C}, v \in V\} \equiv V \oplus iV = \{u + iv | u, v \in V\}$ is an n-dimensional complex vector space. Since distinguishing naturally isomorphic things is a sin I just think of it as the vector space you get when you replace all your real scalars with complex ones. As a vector space over $\mathbb{C}$, H has the same dimension as V. If V has inner product $\langle \cdot, \cdot \rangle_R$ then its complexification has one obtained by extending sesquilinearly[11]. For $\alpha, \beta \in \mathbb{C}, u, v, w \in V, \psi, \phi \in H$

$$\langle u, \alpha v + \beta u \rangle_C = \alpha \langle u, v \rangle_R + \beta \langle u, w \rangle_R \tag{126}$$

$$\langle \phi, \psi \rangle_C = \langle \psi, \phi \rangle_C^* \tag{127}$$

13.2.1 Complexification of 3D space

Let V be a real 3d space with dot product and H its complexification. Let $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ be an orthonormal basis for V. The spherical basis[12]

$$\mathbf{e}_\pm = \frac{1}{\sqrt{2}}(\mathbf{e}_x \mp i\mathbf{e}_y) \qquad \mathbf{e}_0 = \mathbf{e}_z \tag{128}$$

is an orthonormal basis for H wrt $\mathbf{u}^* \cdot \mathbf{v}$. The spherical and cartesian components of a vector $\mathbf{A} = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z = A_-\mathbf{e}_- + A_0\mathbf{e}_0 + A_+\mathbf{e}_+$ are given by

$$A_\pm = \mathbf{e}_\pm^* \cdot \mathbf{A} = \frac{1}{\sqrt{2}}\big(A_x \pm iA_y\big) \qquad A_0 = A_\pm = \mathbf{e}_0^* \cdot \mathbf{A} = A_z \tag{129}$$

$$A_x = \frac{1}{\sqrt{2}}\big(A_+ + A_-\big) \qquad A_y = -\frac{i}{\sqrt{2}}\big(A_+ - A_-\big) \tag{130}$$

bras and kets... H to H* with dagger op... tensor products

Identify all the isomorphic things! Refusing to do so is the cardinal sin of applied math... double duals and original space, dual of tensor products and tensor product of duals, tensor product of space with dual and linear ops on space,...

hermitian ops stretch space by some amount (given by spectrum) in each direction (basis of eigenvectors)

unitary ops like rotations

[11] didn't even make up that word

[12] correct this section! raising and lowering ops are spherical basis vectors, not coefficients of them!

13.3 tensor/kronecker products and sums

tensor product of vector space with underlying field is isomorphic to original vector space so just identify them.

$\exp(A)\hat{\otimes}\exp(B) = \exp\big(A\hat{\oplus}B\big)$

For $i = 1, 2$, let H_i be a Hilbert space with bases $\left|\psi_k^{(1)}\right\rangle$ and $\left|\psi_l^{(2)}\right\rangle$ respectively. Let A